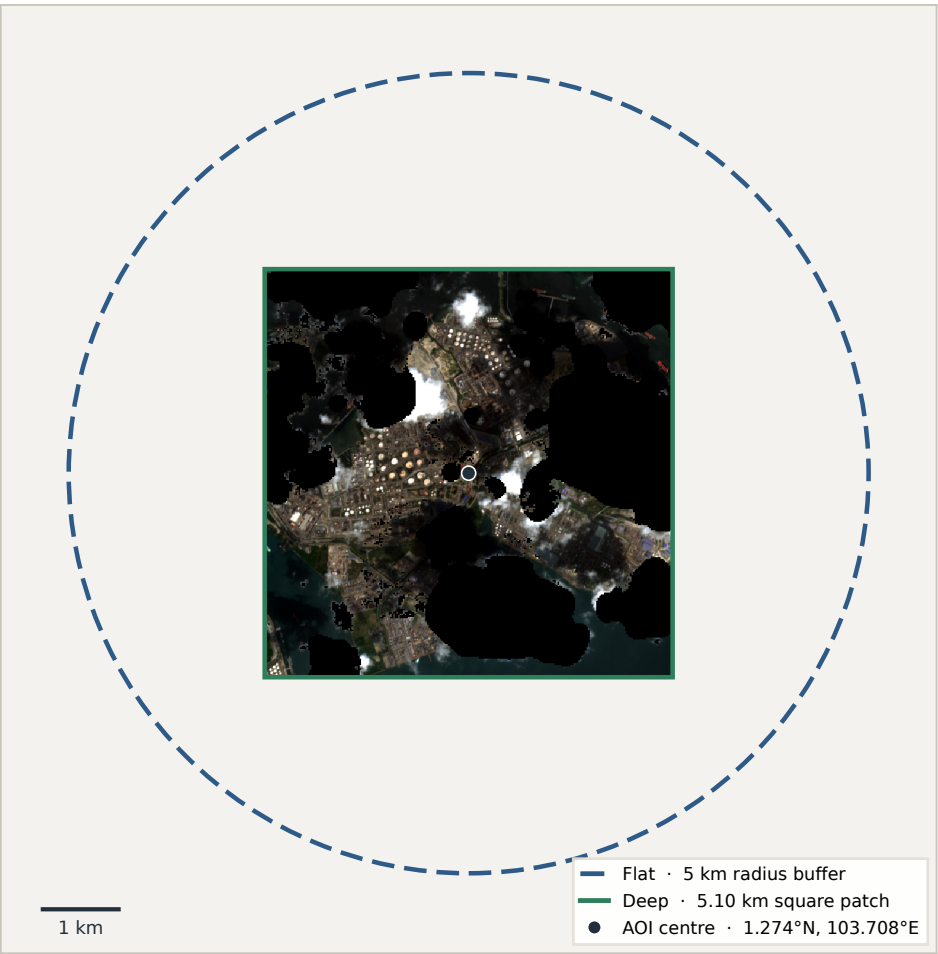


Worked example · Jurong Island (P004) at Friday 14 March 2025

Same origin as Figure 3.7. Brent \$71.94/bbl. Values are the latest eligible observations under the locked lags, not the calendar week of the origin.

(A) Two spatial supports on the January 2025 composite



(B) Remote sensing at this origin

Eligible product: January composite (available from 15 Feb). Carried forward through the four-week window.

Flat · five scalars from the 5 km buffer

NDVI anomaly	+0.56
NDWI anomaly	−0.40
NDBI anomaly	−0.93
BSI anomaly	−0.76
NTL anomaly	+0.36

Deep · frozen 1024-d embedding

Prithvi-EO-2.0-300M · mean-pool
510 px chip · 1 scene · 37% cloud



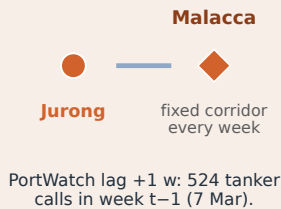
first 48 of 1 024 dimensions (not a forecast)

(C) Shipping neighbourhood · lag +2 weeks

Latest eligible voyage week: 28 Feb 2025. 16 directed lanes · 255 voyages involving Jurong.

Busiest directed lanes

Origin → destination	Voyages
Ningbo → Jurong	57
Jurong → Ningbo	45
Fujairah → Jurong	23
Jurong → Rotterdam	23
Jurong → Fujairah	21
Rotterdam → Jurong	19
+ 10 other lanes (67 voyages)	



(D) Flat ingest · Jurong becomes columns

The five anomalies sit among ~260 weekly features. Because the January composite is carried forward, the four lookback weeks are identical for this site. Shipping series are prefixed columns, not neighbours.

Ridge / XGBoost never see a patch or a graph.

(D) Deep ingest · Jurong stays a place

The 1024-d vector is one of 11 AOI tokens. Temporal then site attention can up-weight Jurong. On the graph it is 1 of 17 nodes, tied to Malacca and to the directed lanes in (C).

Gated fusion later mixes z_{rs} and z_{ship} with finance.